AI for Health

SRIP 2026 — Project Report

Detecting Breathing Irregularities During Sleep Using Physiological Signals

Track: SRIP 2026 Health Sensing Assignment

Dataset: 5-Participant Overnight PSG Pilot Study

Signals: Nasal Airflow (32 Hz), Thoracic Movement (32 Hz), SpO_2 (4 Hz)

Model: SimpleCNN — Single-Branch 1D CNN (36,418 parameters)

Validation: 3-Mode LOPO Cross-Validation (5 folds)

AI Tool Disclosure (Required by Assignment Guidelines)

As required by the assignment guidelines, I explicitly disclose the use of AI assistants (Gemini/Claude) during the development of this project. The AI was used as a technical sounding board to: (1) validate the 1D CNN architecture and the approach of upsampling SpO_2 via polyphase resampling for a unified single-branch input; (2) assist with boilerplate code for Pandas time-series alignment and PyTorch DataLoaders; and (3) structure the LOPO loop to prevent normalisation leakage.

I have thoroughly reviewed, tested, and comprehended every line of code in this repository and am fully prepared to explain all architectural decisions, filtering logic, and metric evaluations during the one-on-one discussion.

Contents

1 Introduction and Data Exploration **2**
- 1.1 Dataset Integrity Audit . . . 2
- 1.2 Parsing and Alignment Challenges . . . 2

2 Signal Preprocessing and Dataset Creation **3**
- 2.1 Digital Signal Processing Pipeline . . . 3
  - 2.1.1 Nasal Airflow and Thoracic Movement (32 Hz) . . . 3
  - 2.1.2 SpO_2 (4 Hz) . . . 3
  - 2.1.3 SpO_2 Resampling: 4 Hz → 32 Hz . . . 3
- 2.2 Window Extraction and Label Assignment . . . 4
  - 2.2.1 Class Consolidation . . . 4
- 2.3 Dataset Summary . . . 4

3 Model Architecture **4**
- 3.1 SimpleCNN . . . 4
- 3.2 Design Rationale . . . 5
- 3.3 Training Configuration . . . 5
- 3.4 Normalisation: Leakage Prevention . . . 5

4 Multi-Mode LOPO Evaluation **6**
- 4.1 Evaluation Protocol . . . 6
- 4.2 Ablation Design: Three Classification Modes . . . 6
- 4.3 Results . . . 6
- 4.4 Analysis . . . 7
  - 4.4.1 Mode: Apnea — Structurally Unlearnable Under LOPO . . . 7
  - 4.4.2 Mode: Hypopnea — Generalizable Features Exist . . . 7
  - 4.4.3 Mode: Event — Correct Clinical Framing . . . 7

5 Limitations and Future Work **7**

6 Conclusion **8**

A Repository Structure **8**

B Breathing Frequency Derivation **9**

C Training Log Evidence for Key Claims **9**

1. Introduction and Data Exploration

The dataset consists of overnight polysomnographic (PSG) recordings from five participants (AP01–AP05) collected across three consecutive nights in May 2024. Three physiological channels are provided per participant, alongside event annotations and sleep staging:

Table 1: Physiological signals, native sampling rates, and clinical roles.

Signal	Native Hz	Final Hz	Clinical Role
Nasal Airflow	32	32	Direct respiratory volume; primary indicator of apnea/hypopnea
Thoracic Mvmt.	32	32	Chest wall expansion; distinguishes obstructive from central events
SpO_2	4	32	Blood oxygen saturation; lags breathing events by 15–30 s

1.1. Dataset Integrity Audit

A cross-reference of all raw file header timestamps revealed two structural gaps with direct consequences for modelling:

Table 2: Per-participant file availability. Missing files marked with dashes.

ID	Duration	Airflow	Thoracic	Events	Sleep Profile
AP01	6.6 h	✓	✓	✓	✓
AP02	8.1 h	✓	✓	✓	—
AP03	7.1 h	✓	✓	—	✓
AP04	7.6 h	✓	✓	✓	✓
AP05	7.4 h	✓	✓	✓	✓

AP03 Critical Consequence for Evaluation

Because AP03 has no events file, every window generated from AP03 is labelled **Normal** by construction. When AP03 is the held-out test fold in LOPO CV, a model that predicts "Normal" for every input achieves near-perfect accuracy. This is confirmed in the training log: Fold 3 validation F1 never exceeded 0.017 across all 50 training epochs in any mode. Fold 3 accuracy must **not** be interpreted as evidence of learning.

1.2. Parsing and Alignment Challenges

The raw files exhibited two structural inconsistencies requiring robust handling in `parsers.py`:

1. **Two distinct timestamp formats** produced by different PSG software export settings:
   - Format A (24-hour): `30-05-2024 21:22:45`
   - Format B (12-hour AM/PM): `5/28/2024 9:28:07 PM`

   Both are resolved by a regex pipeline with `datetime` fallback logic, normalising all timestamps to a seconds-from-recording-start axis.
2. **Differing sampling rates.** All three signals carry absolute datetime timestamps in their data rows. Alignment is performed by intersecting the valid time ranges across all three signals after timestamp parsing, establishing a common time axis.

2. Signal Preprocessing and Dataset Creation

2.1. Digital Signal Processing Pipeline

Human breathing occurs at 10–24 breaths per minute (BrPM), corresponding to 0.17–0.40 Hz. Filters are applied per signal type:

2.1.1. Nasal Airflow and Thoracic Movement (32 Hz)

A 4th-order Butterworth **bandpass filter** (0.17–0.40 Hz) is applied via SciPy's `filtfilt` (zero-phase, forward+backward pass):

- **High-pass component** (0.17 Hz cutoff): removes slow baseline wander from posture changes or sensor drift.
- **Low-pass component** (0.40 Hz cutoff): removes high-frequency movement artefacts.
- **Zero-phase via `filtfilt`**: eliminates temporal shift in the filtered signal, keeping event annotation timestamps correctly aligned.
- **4th-order Butterworth**: maximally flat passband (no ripple) preserves the amplitude morphology of breathing cycles.

2.1.2. SpO_2 (4 Hz)

SpO_2 reflects breathing irregularities as slow desaturation events (timescale: 15–30 s). Applying a bandpass filter would distort this morphology. Instead, a 2nd-order Butterworth **low-pass filter** (cutoff 1.0 Hz, Nyquist = 2 Hz) removes sensor jitter while preserving the authentic desaturation curve.

2.1.3. SpO_2 Resampling: 4 Hz → 32 Hz

To construct a unified [batch, 3, 960] input tensor for the single-branch CNN, SpO_2 is upsampled 8× using SciPy's `resample_poly`. Polyphase resampling applies an anti-aliasing FIR filter during the process, better preserving the signal envelope compared to naive linear interpolation.

Upsampling Trade-off: Why Single-Branch Was Chosen

Upsampling introduces interpolated samples containing no new physiological information. The alternative is a multi-branch architecture that processes SpO_2 at its native 4 Hz. The single-branch approach was chosen for simplicity: with only 4 training

participants per fold, reducing architectural complexity lowers overfitting risk. The 36,418-parameter SimpleCNN is sized for this constraint. A multi-branch design is noted as a future improvement.

2.2. Window Extraction and Label Assignment

Filtered signals are segmented into **30-second windows with 50% overlap**:

- Window length: $30\,\text{s} \times 32\,\text{Hz} = \mathbf{960}$ samples
- Step size: $15\,\text{s} \times 32\,\text{Hz} = 480$ samples
- Tensor shape per window: $[3,\ 960]$

Labelling rule. For each window $[t_s,\ t_e]$, compute fractional overlap with every annotated event $[e_s,\ e_e]$:

$$\text{overlap_frac} = \frac{\max\big(0,\ \min(t_e,\ e_e) - \max(t_s,\ e_s)\big)}{t_e - t_s}$$

If any single event achieves overlap_frac > 0.50, the window receives that event's label. Otherwise, the window is labelled **Normal**.

2.2.1. Class Consolidation

- **Body event**: positional annotation, not a breathing irregularity. Discarded at parse time.
- **Mixed Apnea**: 2 instances in AP01 only. Merged into the Apnea class — a 2-sample standalone class is statistically unlearnable.

2.3. Dataset Summary

Table 3: Per-fold window counts for Event mode (from training log). Class 0 = Normal, Class 1 = Any Event.

Fold	Test	Train (Normal / Event)	Train total	Test windows
1	AP01	6,357 / 621	6,978	1,822
2	AP02	6,467 / 564	7,031	1,769
3	AP03	6,410 / 694	7,104	1,696*
4	AP04	6,315 / 553	6,868	1,932
5	AP05	6,819 / 400	7,219	1,581

*AP03 test set is 100% Normal by construction — see Section 1.

3. Model Architecture

3.1. SimpleCNN

A shallow single-branch 1D CNN was selected over deeper or multi-branch designs. With only 4 training participants per fold, the primary risk is memorisation. The architecture is explicitly sized for the dataset scale at 36,418 parameters.

Input: [Batch, 3, 960] (nasal airflow, thoracic, $SpO_{2\,upsampled}$)

→ `ConvBlock`(3→32, k=7, pool=4) ⇒ [B, 32, 240]
→ `ConvBlock`(32→64, k=5, pool=4) ⇒ [B, 64, 60]
→ `ConvBlock`(64→128, k=3, pool=4) ⇒ [B, 128, 15]

→ `GlobalAvgPool1d` ⇒ [B, 128]
→ `Dropout`(p=0.5)
→ `Linear`(128 → 2)

ConvBlock = Conv1d → BatchNorm1d → ReLU → MaxPool1d(4)
Total parameters: 36,418 (verified: notebook cell output)

Figure 1: SimpleCNN architecture. Three progressively deeper ConvBlocks with MaxPool(4) downsampling, followed by Global Average Pooling and a linear classification head.

3.2. Design Rationale

Global Average Pooling (GAP). GAP compresses $[\mathrm{B}, C, L] \to [\mathrm{B}, C]$ by averaging the time dimension. This enforces *translation invariance*: whether a shallow-breathing pattern exists *anywhere* in the 30-second window, not at a fixed position. It also eliminates a $128 \times 15 \times 2 = 3{,}840$-parameter dense layer that would otherwise push total parameters well above what $N = 5$ can support.

MaxPool1d(4) per block. Each pool layer reduces sequence length by 4×, expanding the receptive field to approximately 448 samples (≈14 s at 32 Hz) by the final block — appropriate for detecting breathing cycle patterns within a 30-second window.

3.3. Training Configuration

Table 4: Hyperparameters applied uniformly across all modes and folds.

Hyperparameter	Value
Optimiser	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight_decay= 10^{-4})
Learning rate	1×10^{-3}
Max epochs	50; early stopping patience = 15 on validation loss
Batch size	32
Loss	CrossEntropyLoss with inverse-frequency class weights
Dropout	$p = 0.5$ before classifier head
Random seed	42
Hardware	CUDA GPU (from training log: `Device: cuda`)

Class-weighted loss. Weights $w_c = 1/\mathrm{count}_c$ are computed strictly within each training fold. This prevents the Normal-dominant class (91–95% of windows) from collapsing gradient updates and producing a trivial majority-class predictor.

3.4. Normalisation: Leakage Prevention

Z-score statistics (μ, σ) are computed *only on the 4-participant training fold* per LOPO iteration and applied statically to the held-out test participant. Computing statistics on

the full dataset before splitting would leak the test distribution into training — a standard and often silent failure in medical ML pipelines.

4. Multi-Mode LOPO Evaluation

4.1. Evaluation Protocol

Leave-One-Participant-Out (LOPO) Cross-Validation: in each of 5 folds the model trains on 4 participants and is evaluated on the held-out participant. The model has never seen the test subject's physiological profile during training, simulating deployment on a genuinely new patient.

Metric Reporting: Binary Average — Event Class Only

All Precision, Recall, and F1 scores are computed with `average='binary'` (scikit-learn). This reports metrics **for Class 1 (the event class) only**. It is *not* macro-averaged over both classes. Accuracy is the standard proportion of correct predictions over all windows. The distinction matters: macro-F1 would partially credit Normal-class performance and inflate the apparent score.

4.2. Ablation Design: Three Classification Modes

A severe structural imbalance was identified: ≈93% of Obstructive Apnea events reside in AP05 alone. This is directly confirmed by the training log — when AP05 is held out, only **15 apnea windows** remain across all 4 training participants (`{0: 6819, 1: 15}`). To isolate this skew from genuine model capability, three independent binary modes were evaluated:

Table 5: Binary classification modes. Each mode redefines Class 1.

Mode	Class 1 (positive)	Purpose
Event	Hypopnea OR Apnea (merged)	First-pass binary screener
Hypopnea	Hypopnea only (Apnea excluded)	Isolates evenly distributed event type
Apnea	Apnea only (Hypopnea excluded)	Stress-tests AP05 concentration problem

4.3. Results

Table 6: 5-fold LOPO CV mean metrics. Precision/Recall/F1 are binary-averaged over the event class (Class 1) only. F1 Std reflects inter-fold variance.

Mode	Accuracy	Precision	Recall	F1	F1 Std
Event (Hyp. + Apnea)	55.5%	11.6%	**59.0%**	17.2%	±12.4%
Hypopnea only	62.9%	9.7%	51.7%	14.7%	±9.0%
Apnea only	91.9%	1.9%	18.7%	3.3%	±5.2%

4.4. Analysis

4.4.1. Mode: Apnea — Structurally Unlearnable Under LOPO

The 91.9% accuracy is misleading. When AP05 is the test fold, only 15 apnea training windows exist. The model learns to predict "Normal" for everything, yielding 91.9% accuracy because the test set itself is 91.9% Normal. The 18.7% Recall confirms this: ≈81% of actual apnea events are missed. This is a *data distribution failure*, not an architecture failure — no classifier can reliably identify a class for which it has seen only 15 training examples.

4.4.2. Mode: Hypopnea — Generalizable Features Exist

Hypopnea events are distributed across AP01, AP02, AP04, and AP05. The model encounters meaningful training volume in every fold. Training log evidence of genuine learning: Fold 1 reached validation F1 = 0.133; Fold 2 reached 0.183; Fold 5 (most hypopnea training data) reached 0.239 at epoch 1 — all significantly above the zero baseline. The 51.7% Recall represents real pattern detection, not majority-class prediction.

4.4.3. Mode: Event — Correct Clinical Framing

By merging all breathing irregularities into one positive class, the model achieves **59.0% Recall** — meaning it flags roughly 3 in 5 true breathing events across completely unseen participants. Class-weighted loss correctly drives the model toward this high-Recall regime.

Why High Recall / Low Precision Is Clinically Correct

For a first-pass medical screening tool, the cost asymmetry is clear:

False negative (missed apnea event): undetected patient risk.
False positive (normal flagged as event): physician review that would have been scheduled eventually regardless.

Maximising Recall at the cost of Precision is the correct directional behaviour for this use case. Precision improvement requires a larger, balanced cohort — not a different model.

5. Limitations and Future Work

- **AP02 missing sleep profile.** The `vis.py` script handles this gracefully (sleep stage panel renders as "Not available"). Sleep-stage-stratified analysis is not possible for AP02.
- **AP03 missing events file.** All AP03 windows are labelled Normal by construction. Fold 3 accuracy is inflated in all three modes and must be excluded from generalisation claims.
- **SpO_2 upsampling artefact.** Polyphase 8× upsampling injects interpolated samples. A multi-branch architecture processing SpO_2 at native 4 Hz would eliminate this limitation and is the recommended architectural next step.

- **Cohort size ($N = 5$) is the binding constraint.** With 4 training participants per fold, the model cannot encounter sufficient physiological diversity. These results constitute a feasibility demonstration, not a clinical validation. A minimum of $N \geq 30$ balanced participants would be required for meaningful multi-class generalisation.
- **Future directions.** Temporal architectures (LSTM, Temporal Fusion Transformer); multi-branch CNN retaining native SpO_2 resolution; sleep-stage conditioning as auxiliary input; subject-adaptive fine-tuning for personalised baselines.

6. Conclusion

This project delivers a complete, end-to-end pipeline for detecting breathing irregularities from overnight PSG data. Key contributions:

1. A robust dual-format European datetime parser handling both 24-hour and AM/PM timestamp conventions across heterogeneous PSG device exports.
2. A domain-correct DSP pipeline: zero-phase Butterworth bandpass filtering (0.17–0.40 Hz) for respiratory signals, low-pass filtering for SpO_2, and polyphase upsampling to a unified 32 Hz axis.
3. A SimpleCNN (36,418 parameters) with Global Average Pooling, sized explicitly for $N = 5$ to prioritise generalisation over memorisation.
4. Rigorous LOPO evaluation with honest acknowledgement of the AP03 labelling artefact, the Apnea concentration problem in AP05, and correct metric reporting (binary average over the event class, not macro).
5. A three-mode ablation study separating the Apnea distribution skew from genuine Hypopnea generalisation, culminating in a binary screener achieving **59% Recall** — the correct clinical target for a first-pass anomaly detector operating at $N = 4$ training participants per fold.

The performance ceiling is determined by cohort scale, not algorithmic complexity. A larger, balanced patient cohort is the single most impactful next step.

A. Repository Structure

Listing 1: Project directory layout.

```
Project Root/
|-- Data/
|   |-- AP01/  nasal_airflow.txt  thoracic_movement.txt
|   |          spo2.txt  flow_events.txt  sleep_profile.txt
|   |-- AP02/  (no sleep_profile.txt)
|   |-- AP03/  (no flow_events.txt)
|   |-- AP04/  (all 5 files)
|   `-- AP05/  (all 5 files)
|-- Visualizations/   AP01--AP05 PDFs (generated by vis.py)
|-- Dataset/
|   |-- breathing_dataset.pkl
|   `-- breathing_dataset.csv
|-- models/
|   `-- cnn_model.py          (SimpleCNN, 36,418 params)
```

```
|-- scripts/
|    |-- parsers.py
|    |-- vis.py
|    |-- create_dataset.py
|    `-- train_model.py
|-- src/
|    |-- config.py
|    `-- utils.py
|-- notebooks/
|    `-- AI_for_Health.ipynb
|-- setup_data.py
|-- requirements.txt
`-- report.pdf
```

B. Breathing Frequency Derivation

Normal adult breathing rate is 10–24 BrPM. Converting to Hz:

$$f_{\min} = \frac{10\,\mathrm{BrPM}}{60\,\mathrm{s/min}} \approx 0.167\,\mathrm{Hz}, \qquad f_{\max} = \frac{24\,\mathrm{BrPM}}{60\,\mathrm{s/min}} = 0.400\,\mathrm{Hz}$$

The 4th-order Butterworth filter provides a maximally flat passband with no amplitude ripple in the 0.17–0.40 Hz band, preserving breathing cycle amplitude morphology. `filtfilt` achieves $-80\,\mathrm{dB/decade}$ roll-off outside the passband with zero net phase delay.

C. Training Log Evidence for Key Claims

The following excerpts from `Training_20260305_225359.log` directly support the analysis in Section 4:

Listing 2: Apnea Mode: Fold 5 confirms only 15 apnea training windows.

```
Fold 5/5 -- Test: AP05  [mode=apnea]
Train 6834 {0: 6819, 1: 15} | Test 1413
Ep   1 | TLoss 0.5886 | VLoss 1.8924 | F1 0.0000
Ep  10 | TLoss 0.4675 | VLoss 2.2095 | F1 0.0000
Early stopping at epoch 24
```

Listing 3: Event Mode: Fold 3 (AP03) confirms F1 stuck near zero.

```
Fold 3/5 -- Test: AP03  [mode=event]
Ep   1 | TLoss 0.6735 | VLoss 1.2244 | F1 0.0157
Ep  10 | TLoss 0.6063 | VLoss 0.9759 | F1 0.0161
Ep  20 | TLoss 0.5811 | VLoss 1.0419 | F1 0.0167
Early stopping at epoch 27
```

Listing 4: Hypopnea Mode: Fold 5 confirms genuine learning (F1 0.24).

```
Fold 5/5 -- Test: AP05  [mode=hypopnea]
Train 7204 {0: 6819, 1: 385} | Test 1441
Ep   1 | TLoss 0.7440 | VLoss 0.7166 | F1 0.2393
Early stopping at epoch 18
```